RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF Operational Open Items & Watch Runbook

A single living tracker for the environment's outstanding items — cluster-store loop, the appliance-root expiry audit (vCenter Aug 10, NSX ~Sept 10), vSAN capacity, and housekeeping — each with verified status and next action. No guessing.

v1.2 VMware Cloud Foundation 9.0.1

INTERNAL

VCF Operational Open Items & Watch Runbook

Prepared by: Virtual Control LLC **Date:** July 27, 2026 **Environment:** VMware Cloud Foundation 9.0.1 **Purpose:** A single living tracker for the environment's outstanding items — each with its **verified** status, next action, and how to check it. Only verified facts are stated; unknowns are marked as such.

> **Near-term action distilled:** rotate **vCenter root by Aug 8** (expires Aug 10) and **NSX passwords before ~Sept 10**, confirm the **Aria license** before ~Aug 3, and **plan the vSphere 9.1 upgrade** to end the cluster-store loop. Everything else is watch or cosmetic.

1. Cluster-store (DKVS) reconfigure loop / esxi04 half-joined replica — OPEN (primary)

- **Status (verified 2026-07-27 23:58):** loop active (count ~2,440, live), esxi04 a half-joined replica, **`is_in_alarm: false`**, vCenter fully healthy. Low-severity noise + recurrence risk, **not an outage.**
- **Root cause:** ESX cluster store (DKVS) 9.0.x issue, Broadcom **KB 385346**, fixed in vSphere 9.1. See [[RCA-vpxd-Failure-Cluster-Store-InvalidState-2026-07-27]] and [[RUNBOOK-ESX-Cluster-Store-DKVS-Loop-vpxd-Troubleshooting-2026-07-27]].
- **Action: no safe self-service fix while not-in-alarm** (disable-DKVS workaround is off-label here). **Durable fix = vSphere 9.1 upgrade.**
- **Check:** `clusterAdmin cluster status` (`is_in_alarm`) + `vpxd.log` loop count. The **cluster-store monitor** (scheduled 30 min) flags **CRIT** on alarm or vpxd PID change.

2. Fleet / LCM appliance root password — ✅ RESOLVED 2026-07-27

- `.95` (lcm.lab.local) root **changed** and **max-age disabled** (`chage -M -1` → "Password expires: never"). Verified. The VCF Ops **"LCM Root password expiry"** alert clears on next collection.

3. vCenter root password — SCHEDULED (Aug 8)

- **Expires 2026-08-10** (verified in the VCF Ops Passwords UI). **Change 2026-08-08** via VCF Operations → Fleet Management → **Passwords** → **vCenter** `root` → **UPDATE**. See [[RUNBOOK-VCF-ESXi-vCenter-Root-Password-Change-2026-07-27]].
- **After:** update the cluster-store monitor's `VC_PW` in `clusterstore-monitor.conf`.

4. Appliance-root expiry — audit result & remaining work

Root	Expiry	State
VCF Ops (`.77`)	never	✅ fixed
Fleet/LCM (`.95`)	never	✅ fixed
vCenter	**2026-08-10**	→ item 3
NSX root/admin/audit	**~2026-09-10**	⚠️ **90-day aging, last change ~46 days ago (per `/api/v1/node/users`); rotate before ~Sept 10**
SDDC Manager root	**2027-05-04**	✅ **verified 2026-07-28** via `su -` + `chage -l root` : last change May 4 2026, **365-day** aging, expires **May 4 2027** — no near-term risk. (`vcf` SSH user also OK to 2027-05-04.)

- **NSX rotation:** change admin/root/audit via NSX (UI/API/CLI) before expiry, or extend the frequency; NSX `password_change_frequency` is 90 days. *(Exact expiry computed from `last_password_change=46` days; confirm on the day.)*
- **Reference:** [[RCA-VCF-Appliance-Root-Password-Expiry-2026-07-27]].

5. Aria Automation license — **WATCH (~Aug 3)**

- **Carried from prior campaigns:** the Aria/Automation license is noted to expire **~2026-08-03**. **Verify the exact date** and confirm Automation is **out of evaluation** (likely covered by the VCF cores license's VCF Automation component) before then. *(Exact date to be confirmed — not asserted here.)*

6. vSAN disk-level capacity imbalance — **LOW**

- **Verified:** vSAN health **GREEN**; datastore **66% used / 34% free (1,356 GB)**; but at least one **capacity disk is <30% free** (imbalance). Not urgent.
- **Action:** low priority — a **vSAN proactive rebalance** evens it out, or reclaim more space. Watch.

7. Port-group vSAN/vMotion mapping (esxi01/02/04) — **COSMETIC**

- **Verified:** on esxi01/02/04 the vSAN vmknic sits on `pg-vmotion` (and vice-versa); **all port groups are VLAN 0**, so it rides the same L2 — **harmless**, and why vSAN is healthy. esxi03 is "correct."
- **Action: leave it.** Only worth re-homing if real VLANs are ever introduced — then per-host, in maintenance mode, with a vSAN-health gate between hosts.

8. "vCenter data collection is slow" (VCF Ops alert) — **LIKELY TRANSIENT**

- **Verified:** CRITICAL, started **2026-07-27 22:56** during heavy activity (DRS/sync/API churn + the DKVS loop). Expected to **clear as load settles.**
- **Action:** watch; if it persists after the environment quiets, review the VCF Ops vCenter adapter health.

9. Housekeeping — `deep-blue` VM

- `deep-blue` — user test VM on vSAN; **dispose when testing is done** to reclaim space.
- `collector` — possible empty vSAN namespace stub; **verify by UUID before any action** (never by friendly name). Benign if truly empty; low-level removal isn't worth the risk.

> `aria_1` **is LIVE / in use — do NOT treat it as an orphan** (corrected 2026-07-27). Earlier notes that listed it as an empty stub were wrong.

10. `aria` CPU co-stop → HA restart — ✅ RESOLVED 2026-07-28

- **What happened:** the `aria` VCF Automation appliance (running **24 vCPU**) soft-locked and was **restarted by HA** — `watchdog: BUG: soft lockup` across CPUs + `Failed to start kube-apiserver.service`.
- **Root cause (verified from live metrics): CPU co-stop** — 24 vCPUs cannot be co-scheduled on the contended nested host. `cpu.costop` spiked **42,954 ms** (chronic ~1,800 ms), `cpu.ready` spiked 32,659 ms. Not a memory or storage fault (balloon 0; CNS volumes stayed Accessible).
- **Fix applied:** graceful guest shutdown → **reduce 24 → 16 vCPU** → power on. Post-fix co-stop **avg 46 ms / max 1,526 ms** (~40× lower); heartbeat green, stable.
- **Guidance:** keep management appliances **≤16 vCPU** on this nested lab; monitor **co-stop**, not host %CPU. Full detail: [[RCA-Aria-VCFA-CPU-CoStop-SoftLockup-HA-Restart-2026-07-28]].

11. CNS / First-Class-Disk "zombie" reclaim — ✅ CLOSED 2026-07-28 (nothing to reclaim)

- **Question:** RVTools flagged **99** `fcd/*.vmdk` as "possibly zombie" — reclaimable junk?
- **Verified answer: NO.** All 99 are **registered, live CNS volumes** (cross-checked against `VStorageObjectManager.ListVStorageObject` → **0 orphans**), and all show **Accessible / Compliant** in the vCenter **Container Volumes** page. RVTools flags *every* FCD as "zombie" because FCDs are never VM-attached — a **false positive**.
- **Disposition: do not delete.** Any real reclaim would have to happen at the K8s/CNS layer (released/unbound PVCs) — none identified. Thread closed.

This is a living document — update status as items are resolved. Resolved 2026-07-27 (not listed above): SDDC host-inventory desync, esxi01 Management-tag guardrail, fleet NTP hardening, `.77` & `.95` appliance roots, DRS returned to Conservative, 9 stuck sync tasks. Resolved/closed 2026-07-28: SDDC Manager root expiry verified (item 4), `aria` co-stop → HA restart (item 10), CNS/FCD reclaim (item 11).

Document Information

Field	Value
Document ID	VC-RUNBOOK-VCF-OPEN-ITEMS-20260727
Version	1.2
Issued	July 27, 2026 (rev. July 28, 2026)
Author	Virtual Control LLC
Classification	Internal
Platform	VMware Cloud Foundation 9.0.1

Revision History

Version	Date	Notes
1.0	2026-07-27	Initial consolidated open-items tracker (items 1-9) on verified data, incl. the appliance-root expiry audit.
1.1	2026-07-27	Corrected item 9: aria_1 is LIVE / in use, NOT an orphan stub (per operator). Removed it from housekeeping; collector to be verified by UUID before any action.
1.2	2026-07-28	Item 4: SDDC Manager root expiry verified (May 4 2027, 365-day). Added item 10 (aria co-stop → HA restart, RESOLVED) and item 11 (CNS/FCD reclaim, CLOSED — nothing to reclaim).

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